

Predicting Frequent Mental Distress and Mental Health Days Using BRFSS 2024

Final Academic Report

Course: STAT 5603: Statistical Learning and Data Mining

Instructor: Professor Peter Califano

Institution: Temple University

Team Members

Bharati Sahani
Enestina Raradza
Prince M. Nkomo
Sheena Huang
Steven Nguyen

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Abstract

Mental health conditions affect roughly one in five American adults annually, yet population-level tools for identifying individuals at risk before a crisis remain limited. This project applied statistical learning methods to the 2024 Behavioral Risk Factor Surveillance System (BRFSS), a nationally administered telephone survey of 457,670 adults, to evaluate whether routinely collected health, demographic, socioeconomic, and behavioral variables can predict mental health outcomes. After selecting 44 candidate variables and removing those with excessive missingness, 27 cleaned predictors were retained for modeling. Two prediction tasks were examined. Scenario A treated frequent mental distress defined as 14 or more mentally unhealthy days in the past 30 as a binary classification problem, tested using LDA, QDA, logistic regression, and random forest. Given a substantial class imbalance, evaluation prioritized recall and precision-recall performance over accuracy. QDA achieved the highest class-1 recall at 0.59, making it the preferred screening model, while random forest achieved the strongest ROC AUC and average precision. Scenario B modeled the number of mentally unhealthy days as a continuous outcome; random forest outperformed linear regression with a test R-squared of 0.33 and RMSE of 6.84 days. Across both scenarios, depression history, cognitive difficulty, physical health burden, and functional limitations were the most consistent predictors, suggesting that mental distress is embedded within broader patterns of physical and socioeconomic vulnerability.

1. Introduction

Mental health is not only a medical concern, but it also affects everyday life including work, family responsibilities, social participation, health care access, and quality of life (World Health Organization, 2022). Instead of only focusing on diagnoses, survey measures such as mentally unhealthy days are useful because they capture recently lived experiences rather than only formal diagnoses. The Behavioral Risk Factor Surveillance System (BRFSS) is designed for this kind of population surveillance, it is a state-based system of health-related telephone surveys conducted among noninstitutionalized adults in the United States and participating areas (Centers for Disease Control and Prevention [CDC], 2025b). The 2024 BRFSS dataset is especially valuable since it is large and covers many factors related to mental health by providing a large, complex, policy-relevant setting for studying mental health patterns across health status, access to care, disability, socioeconomic, and behavioral variables.

The main outcome of interest in this project emanates from a BRFSS mental health question, which asks respondents how many days during the past 30 days their mental health was not good. The analysis considers this outcome in two forms. Firstly, frequent mental distress is treated as a classification problem, where respondents reporting 14 or more poor mental health days are classified as positive cases. This threshold is consistent with the public health definition of frequent mental distress as feeling emotionally unhealthy for at least 14 of the past 30 days (CDC, 2025c). Secondly, the analysis treats the number of poor mental health days as a continuous regression target, which allows the model to estimate overall severity rather than only crossing a binary threshold.

For this project, machine learning is appropriate as mental health is unlikely to be explained by a single variable or a linear relationship. The analysis includes physical health, cost barriers, exercise, depression diagnosis, marital status, education, housing, employment, income, disability measures, age group, insurance coverage, smoking, and heavy drinking. These predictors may interact in many ways that traditional descriptive summaries cannot fully capture. Statistical learning methods allow the project to compare simpler and more interpretable models, such as logistic and linear regression, with more flexible methods, such as random forest and QDA, while assessing performance on held-out test data.

The core research question is, "To what extent can BRFSS 2024 survey variables predict frequent mental distress and the number of mentally unhealthy days among adults, and which predictors most consistently explain these outcomes?"

2. Data

The data came from CDC's 2024 BRFSS annual data file. The CDC documents report 457,670 records in the 2024 public-use file, which matches the raw dataset loaded in the project HTML output (CDC, 2025a). From the data each row represents one adult survey respondent, and each column represents a survey response or calculated variable. The data

set originally included 301 variables, the project narrowed this down to 44 variables that were most relevant to mental health. The variables were selected based on theoretical relevance to mental health, health status, care access, demographic characteristics, functional limitations, and risk behaviors.

The candidate variables that were selected included the target outcome mental health (MENTHLTH), then the predictors: physical health days, sex, medical cost barriers, exercise, depression diagnosis history, marital status, education, home ownership or renting, veteran status, employment, number of children, income, pregnancy status, weight, height, disability indicators, race, age group, urban or rural residence, health plan coverage, smoking status, heavy drinking, and several social or adverse-experience variables.

Variables such as pregnancy status, cognitive memory items with sparse availability, adverse childhood experience variables, social determinants variables, marijuana use, sexual orientation, life satisfaction, loneliness, utility difficulty, and neighborhood safety were eliminated because they had too many missing values, i.e., more than 200,000. After this step, 27 variables remained for analysis.

The data required re-coding of the special survey codes found in BRFSS. For example, on the day-count variables, values such as 88 were re-coded as zero because they represented none. Other codes, such as 'Not sure' and 'refused' were treated as missing and replaced with median values for the physical and mental health day variables. Categorical variables were cleaned similarly, where invalid responses such as 7, 9, 77, and 99 were replaced by the most common valid category (mode). Many variables were also re-coded into simpler formats, such as binary (0/1) indicators. For example, binary yes/no variables such as MEDCOST1, EXERANY2, ADDEPEV3, VETERAN3, DEAF, BLIND, DECIDE, DIFFWALK, DIFFDRES, DIFFALON, health plan status, and heavy drinking were recoded so that the absence of the condition or behavior was represented as zero and the presence as one. Sex was re-coded from the original male/female coding into a 0/1 format. Education and income were collapsed or re-coded into ordered numeric categories with defined baselines.

Continuous variables like height and weight also needed cleaning due to unrealistic values for example, height being above 9000ft. Such outliers were removed from the dataset. The missing values were replaced with medians. Height was converted from the original feet-inches code into a more readable numeric form. Although BMI was calculated during exploratory analysis, it was eventually dropped because it was not strongly related to mental health in this dataset. The process also included checking for duplicate rows and deleting them.

Two modeling datasets were constructed. Scenario A created a binary target named frequent distress, which was coded into 1 and 0 where 1 represented the number of days above 14 days on the MENTHLTH variable, and 0 when less than 14. The MENTHLTH column was then omitted from the predictors to prevent leakage. Scenario B kept MENTHLTH as the continuous outcome variable and to model the number of mentally unhealthy days directly.

Figure 1 summarizes the observed bivariate relationships among the continuous variables used in exploratory analysis. The heatmap highlights that physical health days had the strongest positive relationship with mental health days among the continuous features shown, while the remaining continuous variables showed weaker associations.

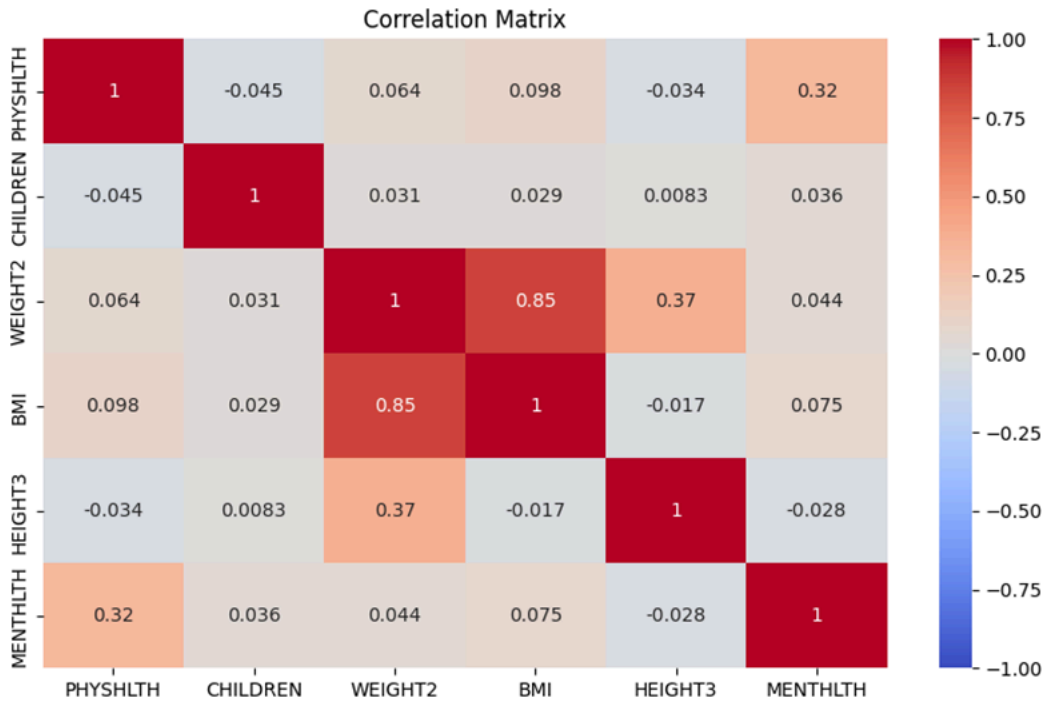


Figure 1. Correlation matrix for selected continuous variables from the HTML output.

Figure 2 goes along with Figure 1. The jittered scatterplot confirms that the association between physical health days and mental health days is visible but noisy. This matters analytically because it suggests that physical health is relevant, but it does not fully explain mental distress, supporting the use of more complex models. The wide scatter supports the use of multivariable statistical learning rather than a one predictor model.

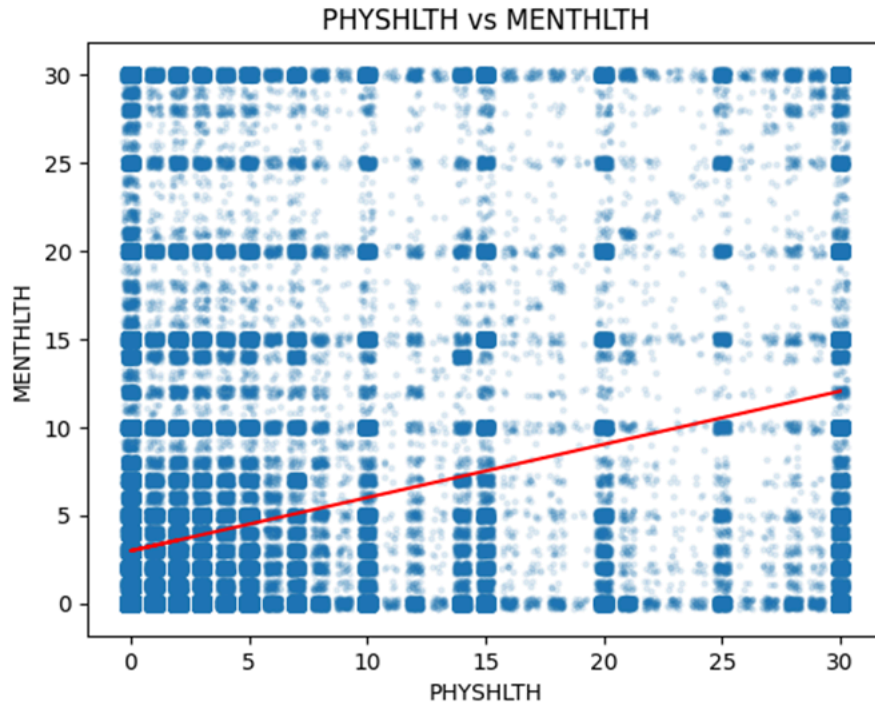


Figure 2. Jittered scatterplot of PHYSHLTH versus MENTHLTH from the HTML output.

3. Methods

The project used two main approaches. Scenario A focused on classification predicting whether someone experiences frequent mental distress. In A the problem was framed as a binary classification task predicting whether someone experiences frequent mental distress. Scenario B focused on regression, predicting the number of unhealthy mental health days. The two approaches are useful because the classification model is aligned with screening and prioritization of care, while the regression model preserves information about severity across the full 0 to 30 day range.

For Scenario A, the data was split into training and test sets using stratified sampling to maintain class balance with random seed 42. The models tested included linear discriminant analysis (LDA), quadratic discriminant analysis QDA, logistic regression, and random forest. Logistic regression provided a simple, interpretable baseline. While random forest was included because it can capture nonlinear relationships and interactions through ensembles of decision trees (Breiman, 2001).

LDA and QDA were included to compare different assumptions about how the data is distributed. LDA assumes shared covariance across classes, while QDA allows the covariance structure to differ by class. This difference is important in this project because mental distress may not vary across predictors in the same way for respondents with and without frequent distress. The implementation used scikit-learn, a widely used Python machine learning library (Pedregosa et al., 2011).

Because Scenario A was imbalanced, accuracy was not the main evaluation metric. The positive class represented only 62,315 of 457,668 observations, so a model could appear accurate by predicting the majority class too often. Instead, the analysis focused on recall, precision, F1 score, ROC AUC, and average precision. Recall was especially important since the goal is similar to screening missing true cases of distress would be more problematic than identifying some false positives. However precision and average precision were still considered because positive predictions could not be reliably interpreted if they contained too many false positives.

In Scenario A, the project also evaluated PCA prior to QDA as part of model refinement. PCA was fitted only on the standardized training predictors and then applied to the test predictors. Two selection criteria were considered: retaining enough principal components to explain 90% of the cumulative variance and retaining components with eigenvalues greater than one. This step served as a diagnostic because QDA can be sensitive to high-dimensional covariance estimation. However, the PCA-based models were not selected as final models, as they resulted in lower class 1 recall compared with the original QDA model.

For Scenario B, the project applied an 80/20 train test split with a random seed of 42. A linear regression model was first used as a baseline, with predictors standardized and evaluated using five-fold cross-validation on the training data. A random forest regression model was then evaluated using three-fold cross-validation, with negative RMSE as the scoring metric. The tuning process considered maximum depth, minimum samples per leaf, and minimum samples required for splitting. Model performance was assessed using test R-squared and RMSE. R-squared captured the proportion of variance explained by the model, while RMSE measured the average prediction error on the scale of mentally unhealthy days. RMSE was particularly informative because the outcome is measured in days, making prediction errors directly interpretable.

4. Results

4.1 Scenario A: Classification of frequent mental distress

Scenario A highlighted the trade-off between screening sensitivity and overall ranking performance. LDA had greater overall accuracy than QDA, but it recognized fewer positive cases. Class 1 recall was 0.422 for LDA and 0.586 for QDA. As the initiative targeted detecting respondents in frequent distress, QDA was more in line with the screening goal, although having lower overall accuracy. The QDA confusion matrix in Figure 3 illustrates this tradeoff. QDA accurately recognized 9,130 positive cases while producing 15,273 false positives.

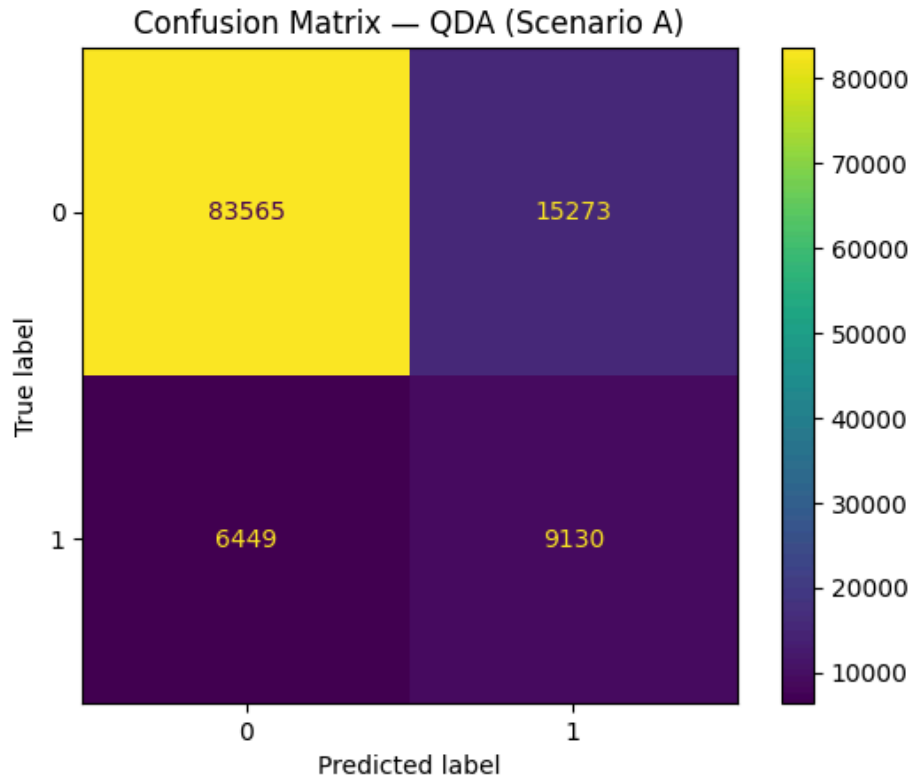


Figure 3. Confusion matrix for QDA in Scenario A from the HTML output.

Model	Average Precision	ROC AUC	Precision, class 1	Recall, class 1	F1, class 1
QDA	0.399	0.803	0.374	0.586	0.457
Logistic Regression	0.516	0.837	0.650	0.307	0.417
Random Forest	0.529	0.846	0.666	0.298	0.412

Table 1. Scenario A model performance comparison on the held-out test set.

Table 1 explains why the ultimate interpretation could not be based on a single parameter. Random forest had the highest average accuracy and ROC AUC, which suggests it evaluated positive cases more accurately across probability thresholds. Logistic regression also had a higher precision than QDA. However, both logistic regression and random forest had class 1 recall near 0.30, indicating that they missed almost seven out of ten positive occurrences at the default classification level. QDA exhibited lower precision but significantly higher recall, making it the most effective model for identifying more respondents who fit the frequent-distress criterion.

Figure 4 illustrates the precision-recall tradeoff identified in Table 1. The three precision-recall curves show that random forest and logistic regression provided higher average accuracy, however QDA seemed more beneficial when the goal was sensitivity to the positive class.

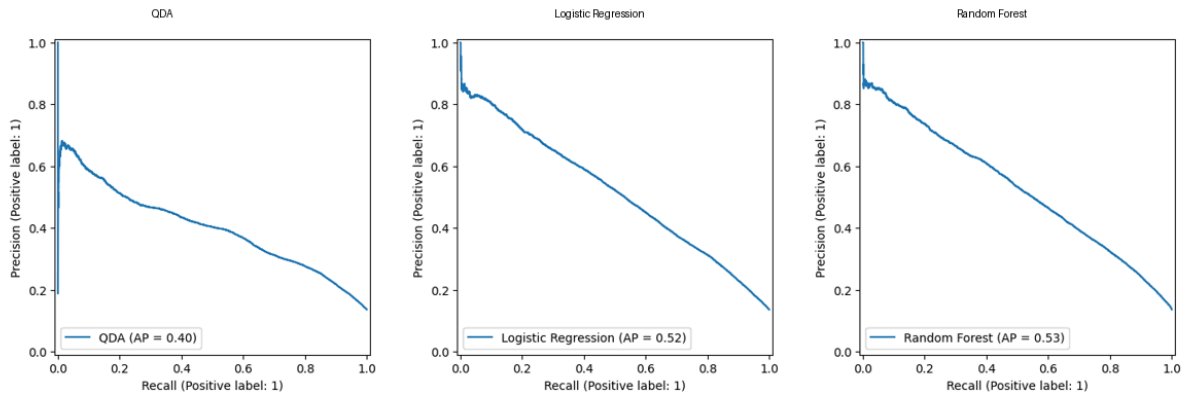


Figure 4. Precision-recall curves for QDA, logistic regression, and random forest from the HTML output.

4.2 Scenario B: Regression of mentally unhealthy days

Scenario B explicitly modeled MENTHLTH as a continuous 0 to 30 day outcome. The linear regression baseline yielded an average cross-validated training R-squared of 0.3055 and a test R-squared of 0.3048. Random forest regression increased test performance, yielding a test R-squared of 0.3279 and an RMSE of 6.8428. The improvement was modest but significant, indicating that the link between predictors and mental health days has a nonlinear or interaction structure that a linear model cannot fully represent.

Model	Test R-squared	Test RMSE
Linear Regression	0.3048	6.9590
Random Forest Regression	0.3279	6.8428

Table 2. Scenario B regression model performance on the held-out test set.

The absolute predictive performance was moderate, not strong. An RMSE of approximately 6.84 days indicates that even the better model frequently underestimated the true number of mentally ill days by a substantial amount. This is not surprising considering that unobserved factors such as recent stressors, social support, treatment access, work circumstances, and prior mental health history all have an impact on self-reported mental health days and were not fully captured in the cleaned feature set.

4.3 PCA iteration

PCA was used to further optimize and tweak the QDA. However, the PCA iteration did not improve the categorization goal. Two concepts were applied, the 90% cumulative variance and Eigenvalue greater than 1 to see which concept gained more class 1 recall. Using 21 main components to explain 90% of the variance yielded a class 1 recall of 0.573, somewhat lower than the original QDA recall of 0.586. The eigenvalue-greater-than-one rule kept 8 components while decreasing class 1 recall to 0.388. Figure 5 depicts the cumulative

variance curve, and Table 3 summarizes the relevant comparisons. The findings imply that the original predictor space had class-separating information that was degraded when reduced to principal components. This is conceivable because PCA focuses on variance preservation rather than class separation.

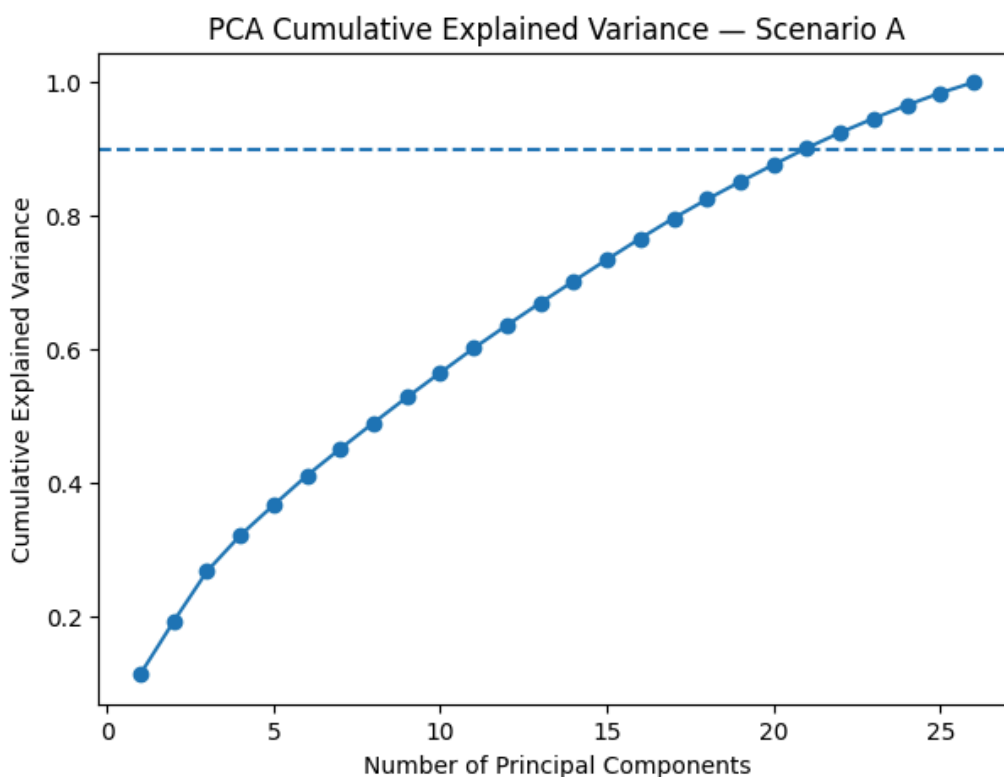


Figure 5. PCA cumulative explained variance for Scenario A from the HTML output.

Model iteration	Predictor representation	Recall, class 1	F1, class 1	Accuracy
Original QDA	Original 26 predictors	0.586	0.457	0.810
PCA + QDA, 90% variance	21 PCs	0.573	0.453	0.811
PCA + QDA, eigenvalue > 1	8 PCs	0.388	0.416	0.852

Table 3. QDA performance before and after PCA feature reduction.

4.4 Feature findings

The final analysis revealed that mental discomfort was substantially associated with previous depression diagnoses, cognitive difficulties, physical health burden, and functional limitations. According to the QDA effect-size comparison, ADDEPEV3, DECIDE, PHYSHLTH, DIFFALON, MEDCOST1, DIFFDRES, and DIFFWALK were the greatest class-separating predictors. These variables represented conceptually meaningful dimensions related to mental health, including diagnosed depression, difficulty concentrating or remembering,

physically unhealthy days, difficulty running errands, cost-related barriers to medical care, difficulty dressing or bathing, and difficulty walking or climbing stairs.

Predictor	Interpretation	QDA absolute standardized difference
ADDEPEV3	Ever diagnosed with depressive disorder	1.0786
DECIDE	Difficulty concentrating or remembering	1.0058
PHYSHLTH	Physically unhealthy days	0.8508
DIFFALON	Difficulty doing errands alone	0.6600
MEDCOST1	Could not see doctor because of cost	0.5562
DIFFDRES	Difficulty dressing or bathing	0.4346
DIFFWALK	Difficulty walking or climbing stairs	0.4187
EMPLOY1	Employment status	0.3127
EXERANY2	Exercise in past 30 days	0.2986
BLIND	Serious difficulty seeing	0.2744

Table 4. Top QDA class-separating predictors from the HTML output.

In Scenario B, the random forest feature importance plot revealed a similar ranking. ADDEPEV3 accounted for roughly 0.28 of impurity significance, DECIDE for approximately 0.18, and PHYSHLTH for approximately 0.15. EXERANY2 was notably smaller, between 0.02 and 0.03. Figure 6 is provided with this interpretation since it shows that the regression model similarly prioritized depression history, cognitive problems, and physical health over all other predictors.

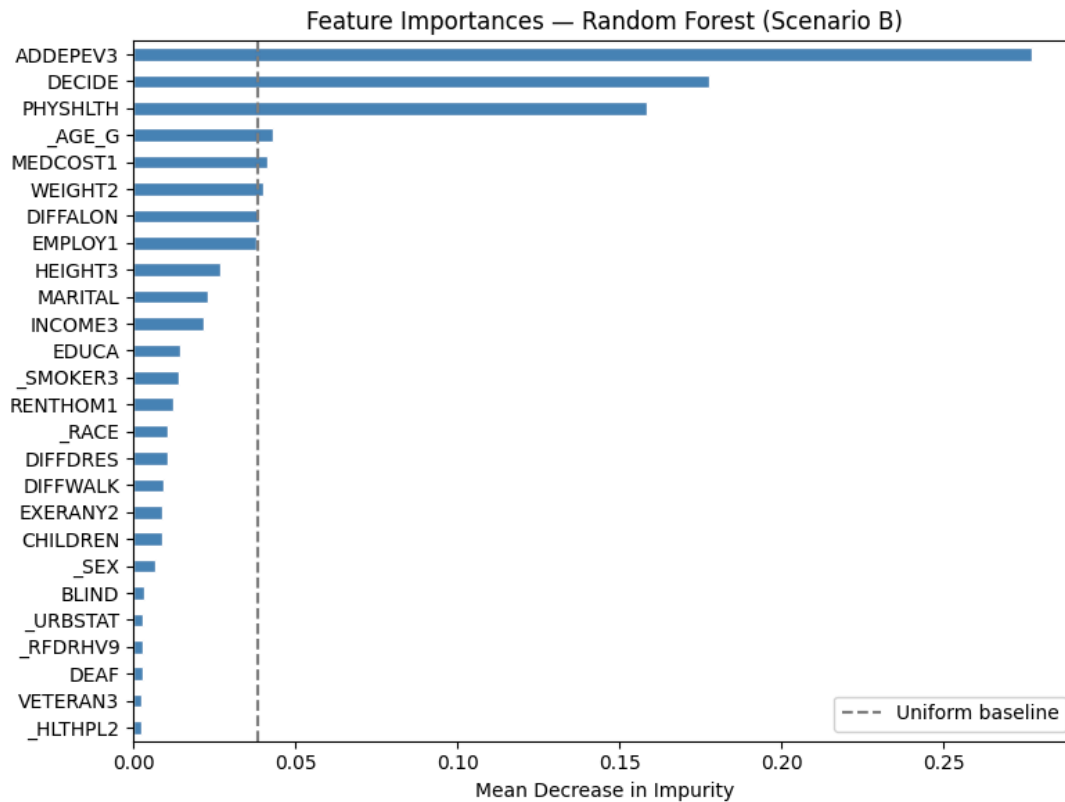


Figure 6. Random forest feature importances for Scenario B from the HTML output.

5. Discussion

As noted by Saito and Rehmsmeier (2015), The main analytical finding is that the most appropriate model is determined by the decision situation. If the goal is to rank respondents based on anticipated risk or to obtain more reliable positive predictions, random forest outperformed the other models since it produced the best ROC AUC and average precision. If the purpose is to screen for regular distress, QDA is more suited because it detects a substantially higher proportion of true positive instances. This distinction is critical for applying health analytics. A technically superior ranking algorithm may nevertheless be ineffective if it misses the majority of the people the intervention is intended to find.

QDA likely outperformed LDA in terms of memory because it enabled the predictor covariance structure to vary between the frequent-distress and non-frequent-distress groups. The class means in the project output demonstrate that the positive group differed significantly on characteristics such as physical health days, depressive diagnosis, trouble concentrating, and functional limitations. A single common covariance matrix may be insufficient to express these disparities. QDA's flexibility assisted in capturing more positives, but it also reduced precision and overall accuracy because the decision boundary identified more respondents as at risk.

The featured findings are substantively consistent. ADDEPEV3 is likely to be the strongest predictor since a prior diagnosis of depressive disorder is strongly associated with mental health outcomes, this is also supported by Kessler et Al (2005). DECIDE is also important since difficulties concentrating or remembering can indicate cognitive load, emotional strain, or other functional disability. PHYSHLTH implies that mental anguish and physical health are linked. The scatterplot demonstrated that this association exists but is not deterministic: many respondents reported poor physical health while having high mental health days, and vice versa. This explains why PHYSHLTH is important but not sufficient on its own.

The significance of functional limitation factors such as DIFFALON, DIFFDRES, and DIFFWALK indicates that mental discomfort may be part of a larger vulnerability pattern. Respondents who struggle with errands, dressing, bathing, walking, or remembering may have decreased freedom, social participation, and access to basic care (Galenkamp et al., 2016). MEDCOST1 also received good marks, showing that cost-related barriers to seeing a doctor are connected with frequent mental issues. This does not evidence that cost constraints cause discomfort, but it does indicate that health care affordability is included in the models' risk profiles.

The regression results should be regarded cautiously. Random forest regression outperformed linear regression, albeit modestly. The last test R-squared value of 0.3279 indicates that the majority of the variation in mental health days remains unexplained. This is not a failure of the analysis, rather, it reflects the complexities of mental health measurement. Timing, recall, stigma, present living circumstances, and unmeasurable clinical and social factors all influence the outcome of a single 30-day self-report. The model can detect trends in the given BRFSS variables, but it is unable to adequately describe individual psychological experiences.

The PCA iteration also offers a useful caution. PCA decreased dimensionality but did not improve the specified classification target. The 90% variance version performed similarly to the original QDA, but with somewhat worse recall. The eigenvalue-greater-than-one version achieved higher accuracy but significantly worse recall. This underscores the distinction between variance preservation and preserving predictive information for an imbalanced target (Li et al., 2018). In this research, keeping the original interpretable predictors was preferable to condensing them into the principal components.

Several constraints must be addressed. First, because BRFSS is a cross-sectional study, the models reflect associations rather than causal effects. Second, many variables are self-reported, which introduces recollection bias, social desirability bias, and variations in how survey questions are interpreted. Third, because the analysis used unweighted modeling, the results should not be taken as nationally representative prevalence estimates. Furthermore, imputation by median or mean preserves sample size while potentially weakening variation in variables with missing or ambiguous data. In addition, several theoretically relevant factors, such as unfavorable childhood experiences and social determinants, were excluded due to significant missingness. Their removal was necessary for model stability, although it is possible that it omitted important mental health predictors. Finally, the default classification

threshold was utilized to compare multiple models. In a practical context, threshold tuning would be required to balance the costs of false negatives and false positives.

6. Conclusion

This project investigated whether BRFSS 2024 survey factors can predict the frequency of mental distress and the number of mentally unwell days, as well as which predictors are most important. The answer is qualified but informative. The models were able to find significant trends, particularly those related to depression history, cognitive difficulty, physical health days, functional limits, and medical cost barriers. QDA was the best appropriate screening-oriented model for the binary frequent-distress task, as it had the highest class 1 recall (0.586). Random forest outperformed linear regression in the regression job, with a test R-squared of 0.3279 and RMSE of 6.8428. These findings indicate that while survey-based health and access variables can predict mental discomfort to some extent, it remains challenging to model properly at the individual level.

The valuable conclusion is that diagnosis history and general demographics should not be the primary factors used to predict mental health risk. Significant prognostic signals are also carried by functional constraints, physical health burden, and care affordability. Using BRFSS survey weights, explicitly adjusting classification thresholds for screening, testing different imbalance-handling techniques, comparing calibrated probabilities, and adding richer social determinant variables if missingness can be handled more carefully are all potential ways to improve the analysis in subsequent work.

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Appendix

Appendix A. Final variable set

Variable	Use in final analysis
SEX	Sex recoded as 0/1.
PHYSHLTH	Number of physically unhealthy days in the past 30 days.
MENTHLTH	Number of mentally unhealthy days in the past 30 days, used as Scenario B target.
MEDCOST1	Could not see a doctor due to cost, recoded binary.
EXERANY2	Exercise in past 30 days, recoded binary.
ADDEPEV3	Ever diagnosed with depressive disorder, recoded binary.
MARITAL	Marital status, cleaned and recoded.
EDUCA	Education level, cleaned and grouped.
RENTHOM1	Own, rent, or other housing arrangement, cleaned and recoded.
VETERAN3	Veteran status, recoded binary.
EMPLOY1	Employment status, cleaned and recoded.
CHILDREN	Number of children in household, cleaned as integer.
INCOME3	Income level, cleaned and recoded.
WEIGHT2	Reported weight in pounds, cleaned and imputed.
HEIGHT3	Reported height, cleaned and converted to readable feet-inches numeric format.
DEAF	Serious difficulty hearing, recoded binary.
BLIND	Serious difficulty seeing, recoded binary.
DECIDE	Difficulty concentrating or remembering, recoded binary.
DIFFWALK	Difficulty walking or climbing stairs, recoded binary.
DIFFDRES	Difficulty dressing or bathing, recoded binary.
DIFFALON	Difficulty doing errands alone, recoded binary.
RACE	Race/ethnicity category, cleaned and recoded.
_AGE_G	Age group, recoded with 65 or older as baseline in the completed analysis.
_URBSTAT	Urban or rural residence, recoded binary.
HLTHPL2	Health plan coverage, recoded binary.
SMOKER3	Smoking status, cleaned and recoded.
RFDRHV9	Heavy drinking indicator, recoded binary.

Appendix B. Generative AI disclosure

Generative AI was used as an assistance tool during the project workflow. The HTML output explicitly notes AI assistance for selected graph coding, random forest adjustment ideas, PCA integration into QDA. For this written report, some AI assistance was used to organize the narrative and polish academic wording.